SentinelForge IR
Incident response & security engineering

Data Source Connection Manual

Step-by-step instructions for sending logs from firewalls, Windows servers, Microsoft 365 / Entra ID, Microsoft Defender and custom systems into SentinelForge IR, and for enabling threat-intelligence lookups.

FortiGate firewall · Windows Security & Sysmon · Microsoft Entra ID · Microsoft Defender XDR · Azure Monitor / Sentinel · Custom JSON API · NepSec Intel

Application
https://sentinelforge.kuinkel.com

Ingestion API
https://sentinelforge-api.kuinkel.com

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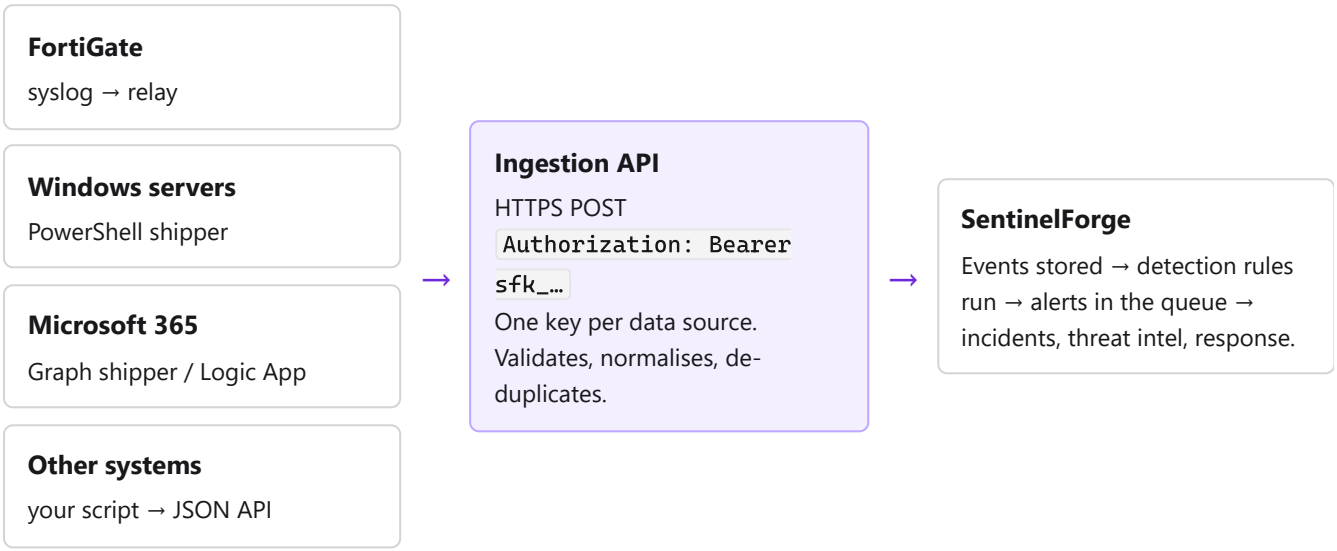
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1. How log collection works

SentinelForge receives logs over HTTPS. Every log source posts batches of events to the ingestion API with its own secret key. The platform normalises the events, stores them, and immediately runs the detection rules over them.



Key terms

Term	Meaning
Data source	A named log stream, for example “FortiGate HQ” or “DC01 Security log”. Each has a health status (Reporting / Stale), an event count and its own ingest key.
Ingest key	A secret starting with `sfk_`. It is shown **once** when created; only its hash is stored. It can only send events: it cannot read data or sign in.
Shipper / relay	A small program that collects logs where they are produced and posts them to the API. SentinelForge provides ready-made ones for FortiGate, Windows and Microsoft Graph (folder `connectors/` in the repository).
Endpoint	The API address a shipper posts to. Each format has its own endpoint (table below).

Ingestion endpoints

Endpoint (on https://sentinelforge-api.kuinkel.com)	Accepts	Used by
`POST /api/v1/ingest/fortigate`	FortiOS syslog lines (key=value or JSON)	FortiGate relay
`POST /api/v1/ingest/microsoft/windows-event`	Windows Security, System, Sysmon events	Windows shipper, Log Analytics
`POST /api/v1/ingest/microsoft/entra-signin`	Entra ID sign-in logs	Graph shipper, Logic App
`POST /api/v1/ingest/microsoft/entra-audit`	Entra ID audit logs	Graph shipper, Logic App

`POST /api/v1/ingest/microsoft/defender-alert`	Defender XDR / Entra ID Protection alerts	Graph shipper, Logic App
`POST /api/v1/ingest/events`	Normalised JSON events (any system)	Your own scripts and tools

What the detections look for

Rule	Detects	Needs logs from
SF-AUTH-001	Five or more failed logins for one account from one address, then a success within 10 minutes	FortiGate SSL-VPN, Windows 4624/4625, Entra ID sign-ins
SF-PROC-001	Suspicious command lines: encoded PowerShell, LOLBins (certutil, mshta, rundll32…), Office spawning shells	Windows 4688 with command line, Sysmon 1
SF-IAM-001	New account added to a privileged group or role within 30 minutes	Windows 4720 + 4728/4732/4756, Entra ID audit
SF-DEF-001	Audit log cleared or logging service stopped	Windows 1102/104, 7036
SF-NET-001	Outbound connections on remote-access ports, large uploads, watchlisted destinations	FortiGate traffic, Sysmon 3
SF-MAIL-001	An emailed link visited by the recipient within an hour	Email gateway + DNS / web filter (FortiGate)
SF-EXT-001	Medium/high alerts from security products	FortiGate IPS/AV, Microsoft Defender

2. Before you start

You need

☐ A SentinelForge **administrator** account (only admins can create data sources and keys; demo accounts cannot).

☐ For each source, a machine that can reach `https://sentinelforge-api.kuinkel.com` on TCP 443 (outbound HTTPS).

☐ The connector scripts from the repository folder `connectors/`:

- ☐ `connectors/fortigate/fortigate_relay.py`
- ☐ `connectors/windows/Send-WindowsEvents.ps1`
- ☐ `connectors/microsoft-graph/ship_graph_logs.py`

☐ Python 3.10 or newer for the Python connectors (no extra packages). Windows PowerShell 5.1 or PowerShell 7 for the Windows shipper.

Check connectivity

From the machine that will run a shipper, the health check must return `"status":"ok"`:

```
# Windows PowerShell
Invoke-RestMethod https://sentinelforge-api.kuinkel.com/api/v1/health

# Linux / macOS
curl -s https://sentinelforge-api.kuinkel.com/api/v1/health
```

If your network uses an outbound proxy, set `HTTPS_PROXY` for the Python connectors (they honour it) or use `-Proxy` with PowerShell.

Handling ingest keys safely

Treat every ingest key like a password.

- Store it in an environment variable or a file readable only by the service account. Never put it on a command line, in a script, a ticket, chat or email.
- Use one key per source so you can revoke one without affecting the others.
- If a key may have leaked,
  Rotate
  it (Settings → Data sources). The old key stops working at once.

Free-tier capacity

The hosted database has 0.5 GB. Security events (logins, alerts, admin changes) are small and always worth sending. High-volume traffic logs are filtered by default in the FortiGate relay; keep the default unless you have a larger database.

3. Create a data source and ingest key

1. Sign in at https://sentinelforge.kuinkel.com with an administrator account.
2. Open **Settings** (left sidebar, under *Govern*) and find the **Data sources** panel. Click **Add source**.
3. Enter a clear **Name** (for example *FortiGate HQ*, *DC01 Security log*, *Entra ID*) and choose the **Type**:

Connecting	Type
FortiGate firewall	`fortigate`
Windows Security / System log	`windows_security`
Sysmon (separate source, optional)	`sysmon`
Entra ID sign-in and audit logs	`microsoft_entra`
Microsoft Defender alerts	`microsoft_defender`
Anything else	closest match, or `custom`

4. Click **Create and show key**. Copy the key (it starts with `sfk_`) into your password manager or straight into the shipper's configuration, then click **I have stored it**. **It cannot be shown again.**

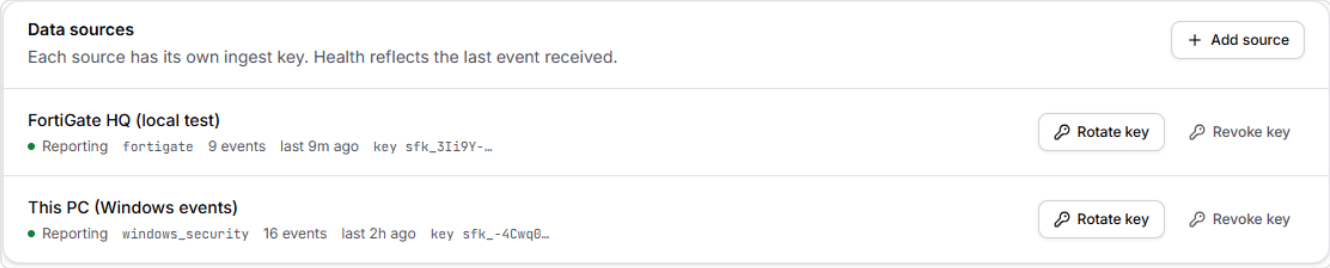


Figure 1 – Settings → Data sources. Each row shows health, type, event count, last event and the first characters of the key.

Managing keys later

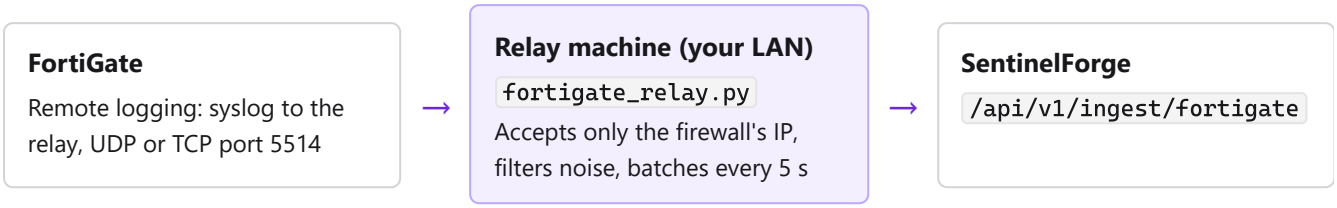
Button	What it does	When to use it
Rotate key	Issues a new key and invalidates the old one immediately. The new key is shown once.	Scheduled rotation, staff changes, suspected exposure. Update the shipper with the new key right away.
Revoke key	Invalidates the key; the source and its events are kept. Shippers using it get `401`.	Decommissioning a log sender, or stopping a misbehaving one.
Issue key	Creates a key for a source whose key was revoked.	Re-enabling a source.
Delete	Removes the source. Only offered when it has **no** stored events, so evidence is never lost.	Cleaning up a source created by mistake.

Removing people instead of keys?

Settings → *Users and roles* → **Remove** signs the person out everywhere and blocks sign-in while keeping their history. Ingest keys are not tied to people, so removing a user does not stop any shipper.

4. FortiGate firewall

FortiGate sends logs only as syslog. A small relay program inside your network receives the syslog and forwards it to SentinelForge over HTTPS.



What is collected

FortiGate log	Becomes	Detection
SSL-VPN login failures and tunnels established	Login failure / success	VPN password guessing (SF-AUTH-001)
Administrator logins to the firewall	Login failure / success	SF-AUTH-001
Accepted traffic sessions (filtered)	Network connection	SF-NET-001
DNS filter, web filter	DNS query, web request	SF-MAIL-001
IPS, antivirus, anomaly, blocked application control	Security alert	SF-EXT-001

4.1 Create the data source

Follow chapter 3 with type `fortigate` (for example “FortiGate HQ”). Keep the `sfk_` key ready.

4.2 Prepare the relay machine

Any always-on machine on the LAN that the FortiGate can reach: a small Linux VM, a Windows server, or a Raspberry Pi. Requirements: Python 3.10+, outbound HTTPS, and inbound **UDP and TCP 5514** from the FortiGate.

Option A – Windows server

1. Install Python 3 from python.org (tick *Add python.exe to PATH*). Copy `fortigate_relay.py` to `C:\SentinelForge\`.
2. Allow the syslog port through Windows Firewall (PowerShell as Administrator; use your FortiGate's IP):

```
New-NetFirewallRule -DisplayName "SentinelForge relay UDP" -Direction Inbound -Protocol
UDP -LocalPort 5514 -RemoteAddress 192.168.1.99 -Action Allow
New-NetFirewallRule -DisplayName "SentinelForge relay TCP" -Direction Inbound -Protocol
TCP -LocalPort 5514 -RemoteAddress 192.168.1.99 -Action Allow
```

3. Store the settings as machine environment variables (not in the script):

```
[Environment]::SetEnvironmentVariable("SENTINELFORGE_URL", "https://sentinelforge-
api.kuinkel.com", "Machine")
[Environment]::SetEnvironmentVariable("SENTINELFORGE_INGEST_KEY",
"sfk_paste_your_key_here", "Machine")
```

4. Test it in a new PowerShell window; it should print *Listening for FortiGate syslog…*:

```
python C:\SentinelForge\fortigate_relay.py --allow 192.168.1.99
```

5. Run it permanently as a startup task (Administrator PowerShell):

```
$action  = New-ScheduledTaskAction -Execute "python.exe" -Argument
"C:\SentinelForge\fortigate_relay.py --allow 192.168.1.99"
$trigger = New-ScheduledTaskTrigger -AtStartup
$settings = New-ScheduledTaskSettingsSet -RestartCount 999 -RestartInterval (New-TimeSpan
-Minutes 1) -ExecutionTimeLimit ([TimeSpan]::Zero)
Register-ScheduledTask -TaskName "SentinelForge FortiGate relay" -Action $action -Trigger
$trigger -Settings $settings -User "SYSTEM" -RunLevel Highest
Start-ScheduledTask -TaskName "SentinelForge FortiGate relay"
```

Option B – Linux (systemd)

```
sudo mkdir -p /opt/sentinelforge /etc/sentinelforge
sudo cp fortigate_relay.py /opt/sentinelforge/
sudo tee /etc/sentinelforge/relay.env >/dev/null <<'EOF'
SENTINELFORGE_URL=https://sentinelforge-api.kuinkel.com
SENTINELFORGE_INGEST_KEY=sfk_paste_your_key_here
EOF
sudo chmod 600 /etc/sentinelforge/relay.env

sudo tee /etc/systemd/system/sentinelforge-relay.service >/dev/null <<'EOF'
[Unit]
Description=SentinelForge FortiGate syslog relay
After=network-online.target
Wants=network-online.target

[Service]
EnvironmentFile=/etc/sentinelforge/relay.env
ExecStart=/usr/bin/python3 /opt/sentinelforge/fortigate_relay.py --allow 192.168.1.99
DynamicUser=yes
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
EOF
sudo systemctl daemon-reload
sudo systemctl enable --now sentinelforge-relay
sudo journalctl -u sentinelforge-relay -f      # watch the log
sudo ufw allow from 192.168.1.99 to any port 5514   # if ufw is enabled
```

Relay options

Option	Meaning
`--allow` required	Comma-separated IPs of your FortiGate(s). Packets from any other address are dropped, so nobody can inject fake firewall logs.
`--port` 5514	UDP and TCP port to listen on. 5514 avoids needing administrator rights for port 514.
`--traffic` interesting	**interesting**: only accepted sessions to public addresses on unusual ports or with more than 10 MiB sent. **all**: every session (very high volume). **none**: no traffic logs. VPN, admin, DNS, web filter and IPS logs are always sent.
`--bind` 0.0.0.0	Local address to listen on; set the LAN IP to listen on one interface only.
`--batch` `--interval`	Up to 200 lines per request, sent every 5 seconds.
`--buffer` 50000	Lines kept in memory while SentinelForge is unreachable; the relay retries automatically.

4.3 Configure the FortiGate

Using the CLI (recommended)

Open the CLI console (top right in the GUI, or SSH) and run, replacing `192.168.1.50` with the relay machine's IP:

```
config log syslogd setting
    set status enable
    set server "192.168.1.50"
    set port 5514
    set mode udp                 # or "reliable" for TCP
    set format default           # "json" is also supported
    set facility local7
end

config log syslogd filter
    set severity information
    set forward-traffic enable
    set local-traffic disable
    set multicast-traffic disable
    set sniffer-traffic disable
    set anomaly enable
    set voip disable
end
```

If the FortiGate sends logs from a management interface or VDOM, pin the source address and put that same IP in the relay's `--allow`:

```
config log syslogd setting
    set source-ip 192.168.1.99
end
```

Using the GUI

1. **Log & Report** → **Log Settings** → *Remote Logging and Archiving*.
2. Enable **Send logs to syslog** and enter the relay's IP address. Click **Apply**.
3. The GUI uses port 514 by default; set port 5514 and the mode with the CLI commands above.

4.4 Make sure the right events are logged

1. **Policy & Objects** → **Firewall Policy**: open each Internet-facing policy and set **Log Allowed Traffic** to *All Sessions* (or at least *Security Events*).
2. **Security Profiles**: in the **DNS Filter**, **Web Filter** and **IPS** profiles used by those policies, enable logging (for DNS filter: *Log all DNS queries and responses*).
3. SSL-VPN logins and administrator logins are logged by default (**Log & Report** → **Log Settings** → **Event Logging**: *VPN activity* and *System activity* must be enabled).

4.5 Confirm

1. On the FortiGate, check that packets leave towards the relay: `diagnose sniffer packet any 'port 5514' 4 5`.
2. The relay prints statistics every 5 minutes, for example `stats received=812, dropped_filter=640, sent=172, …`.
3. In SentinelForge, **Settings** → **Data sources** shows the FortiGate as *Reporting*; **Events** shows its records. Then run the safe test in chapter 10.

5. Windows servers and workstations

The PowerShell shipper runs every 5 minutes on each server (domain controllers first), reads the security-relevant event IDs since its last run, and posts them to SentinelForge.

Events collected

Log	Event IDs	Meaning
Security	4624, 4625	Successful / failed logon (machine accounts and system logons are skipped)
Security	4688	Process created (with command line when enabled)
Security	4720	User account created
Security	4728, 4732, 4756	Member added to a global / local / universal security group
Security	1102	Security log cleared
System	7036, 104	Service stopped (e.g. Windows Event Log, Sysmon); log cleared
Sysmon	1, 3, 22	Process creation with hashes, network connection, DNS query (only if Sysmon is installed)

5.1 Turn on the required auditing

On a domain, set these in a GPO linked to servers and domain controllers: **Computer Configuration** → **Policies** → **Windows Settings** → **Security Settings** → **Advanced Audit Policy Configuration**. For a single machine, run in an Administrator command prompt:

```
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
auditpol /set /subcategory:"Process Creation" /success:enable
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
auditpol /set /subcategory:"Security Group Management" /success:enable
auditpol /set /subcategory:"Audit Policy Change" /success:enable
```

Include command lines in event 4688 (needed for suspicious-process detection). GPO: **Administrative Templates** → **System** → **Audit Process Creation** → **Include command line in process creation events = Enabled**, or on one machine:

```
reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v
ProcessCreationIncludeCmdLine_Enabled /t REG_DWORD /d 1 /f
```

Optional: Sysmon
adds file hashes, network connections and DNS queries per process. Install it from Microsoft Sysinternals with a community configuration (for example SwiftOnSecurity's) — the shipper picks it up automatically.

5.2 Install the shipper

1. Create a data source (chapter 3), type `windows_security`, one per server or one shared by a group of servers.
2. Copy `Send-WindowsEvents.ps1` to `C:\SentinelForge\` on the server.
3. Store the key at machine level (Administrator PowerShell):

```
[Environment]::SetEnvironmentVariable("SENTINELFORGE_INGEST_KEY",
"sfk_paste_your_key_here", "Machine")
```

4. Run it once by hand to test (Administrator PowerShell, because the Security log requires it):

```
powershell -NoProfile -ExecutionPolicy Bypass -File C:\SentinelForge\Send-
WindowsEvents.ps1 -ApiUrl https://sentinelforge-api.kuinkel.com
```

   Expected output: `… sent 214 events, 214 new, 0 alerts raised`. The first run reads the last 60 minutes (`-LookbackMinutes` changes this).

5. Schedule it every 5 minutes as SYSTEM:

```
$action  = New-ScheduledTaskAction -Execute "powershell.exe" -Argument '-NoProfile -
ExecutionPolicy Bypass -File "C:\SentinelForge\Send-WindowsEvents.ps1" -ApiUrl
https://sentinelforge-api.kuinkel.com'
$trigger = New-ScheduledTaskTrigger -Once -At (Get-Date) -RepetitionInterval (New-TimeSpan
-Minutes 5)
Register-ScheduledTask -TaskName "SentinelForge event shipper" -Action $action -Trigger
$trigger -User "SYSTEM" -RunLevel Highest
```

Shipper options

Parameter	Meaning
`-ApiUrl` required	https://sentinelforge-api.kuinkel.com
`-StateFile`	Where the last-sent time is kept (default `C:\ProgramData\SentinelForge\windows-events.state`). Delete it to resend the lookback window; duplicates are ignored.
`-LookbackMinutes` 60	How far back the first run reads.
`-IncludeNetworkLogons`	Also send successful network logons (type 3). Very frequent on servers; failures are always sent.
`-BatchSize` 200	Events per request.

6. Microsoft Entra ID and Microsoft Defender

The Graph shipper reads Entra ID sign-in and audit logs and Defender XDR alerts from Microsoft Graph using an app registration, and posts them to SentinelForge. Run it on any always-on machine with outbound HTTPS.

Stream	Graph source	Permission	Licence
`entra-signin`	/auditLogs/signIns	AuditLog.Read.All, Directory.Read.All	**Entra ID P1 or P2**
`entra-audit`	/auditLogs/directoryAudits	AuditLog.Read.All	Any
`defender-alert`	/security/alerts_v2	SecurityAlert.Read.All	Defender for Endpoint / Office 365 / Identity, or Microsoft 365 E5 / Business Premium

MFA prompts are not failures.
Sign-in records that only asked for MFA, "Stay signed in?" or a password change are skipped, so normal MFA use never looks like password guessing.

6.1 Register the application

1. Open **entra.microsoft.com** as a Global Administrator or Privileged Role Administrator → **Applications** → **App registrations** → **New registration**. Name: *SentinelForge log shipper*; *Accounts in this organizational directory only*; no redirect URI. Click **Register**.
2. On the **Overview** page copy the **Directory (tenant) ID** and **Application (client) ID**.
3. **API permissions** → **Add a permission** → **Microsoft Graph** → **Application permissions**. Add `AuditLog.Read.All`, `Directory.Read.All` and `SecurityAlert.Read.All`. Then click **Grant admin consent for <your tenant>**; every permission must show a green tick.
4. **Certificates & secrets** → **Client secrets** → **New client secret**, expiry 6–12 months. Copy the secret **Value** immediately (it is hidden afterwards). Put a reminder in your calendar to renew it before it expires.

6.2 Create data sources

Create one source of type `microsoft_entra` (sign-ins and audit) — and optionally a separate `microsoft_defender` source if you want separate health tracking. One key can serve all three streams.

6.3 Configure and run the shipper

1. Copy `ship_graph_logs.py` to the machine (e.g. `C:\SentinelForge\` or `/opt/sentinelforge/`).

2. Set the configuration as environment variables for the account that will run it:

```
# Windows (PowerShell); use "Machine" if it will run as SYSTEM
[Environment]::SetEnvironmentVariable("MS_TENANT_ID", "<tenant id>", "User")
[Environment]::SetEnvironmentVariable("MS_CLIENT_ID", "<client id>", "User")
[Environment]::SetEnvironmentVariable("MS_CLIENT_SECRET", "<secret value>", "User")
[Environment]::SetEnvironmentVariable("SENTINELFORGE_URL", "https://sentinelforge-
api.kuinkel.com", "User")
[Environment]::SetEnvironmentVariable("SENTINELFORGE_INGEST_KEY",
"sfk_paste_your_key_here", "User")

# Linux: put the same five lines (NAME=value) in /etc/sentinelforge/graph.env, chmod 600
```

3. Run once to test:

```
python ship_graph_logs.py
# without an Entra ID P1/P2 licence:
python ship_graph_logs.py --streams entra-audit,defender-alert
```

Expected output, one line per stream: `entra-signin: 120 fetched, 120 new, 0 duplicate, 3 skipped, 0 alerts`.

4. Run it continuously or on a schedule:

```
# keep running, one pass every 5 minutes
python ship_graph_logs.py --loop 300

# or Windows Task Scheduler every 5 minutes
$action  = New-ScheduledTaskAction -Execute "python.exe" -Argument
"C:\SentinelForge\ship_graph_logs.py"
$trigger = New-ScheduledTaskTrigger -Once -At (Get-Date) -RepetitionInterval (New-TimeSpan
-Minutes 5)
Register-ScheduledTask -TaskName "SentinelForge Graph shipper" -Action $action -Trigger
$trigger

# or Linux cron (as the user that owns graph.env)
*/5 * * * * set -a; . /etc/sentinelforge/graph.env; /usr/bin/python3
/opt/sentinelforge/ship_graph_logs.py
```

Shipper options

Option	Meaning
`--streams`	Comma-separated: `entra-signin`, `entra-audit`, `defender-alert` (default all).
`--lookback` `60`	Minutes fetched on the first run.
`--state`	Checkpoint file (default next to the script). Each stream continues from its last record.
`--loop`	Seconds between passes; 0 runs once.

The shipper waits and retries automatically when Microsoft Graph throttles (HTTP 429). Microsoft makes new sign-in records available after a few minutes; that delay is normal.

7. Azure Monitor, Log Analytics and Microsoft Sentinel

If your logs already flow into a Log Analytics workspace (Microsoft Sentinel, Defender for Cloud, Azure Monitor Agent), you can forward them without running anything on your servers, using a Logic App.

7.1 Store the ingest key in Key Vault

1. Create a data source in SentinelForge (chapter 3) and copy the key.
2. In the Azure portal open (or create) a **Key Vault** → **Secrets** → **Generate/Import**, name `sentinelforge-ingest-key`, value = the key.

7.2 Build the Logic App

1. Create a **Logic App (Consumption)**. Under **Identity** enable the system-assigned managed identity, and give it **Log Analytics Reader** on the workspace and **Key Vault Secrets User** on the vault.
2. Trigger: **Recurrence**, every 5 minutes.
3. Action: **Azure Monitor Logs** → **Run query and list results** on your workspace with one of these queries (time range: last 5 minutes):

```
// Windows security events  → endpoint .../ingest/microsoft/windows-event
SecurityEvent
| where EventID in (4624, 4625, 4688, 4720, 4728, 4732, 4756, 1102)
| project TimeGenerated, Computer, EventID, EventData

// Entra ID sign-ins → .../ingest/microsoft/entra-signin
SigninLogs
| project TimeGenerated, UserPrincipalName, IPAddress, ResultType, ResultDescription,
AppDisplayName, LocationDetails, DeviceDetail

// Entra ID audit → .../ingest/microsoft/entra-audit
AuditLogs
| where OperationName in ("Add user", "Add member to role", "Add member to group", "Add
eligible member to role")
| project TimeGenerated, OperationName, Result, InitiatedBy, TargetResources

// Security alerts → .../ingest/microsoft/defender-alert
SecurityAlert
| project TimeGenerated, AlertName, AlertSeverity, ProviderName, CompromisedEntity,
Description
```

4. Action: **Azure Key Vault** → **Get secret** `sentinelforge-ingest-key`. In its *Settings*, turn on **Secure outputs** so the key never appears in run history.

5 Action: **HTTP**

Method	**POST**
URI	`https://sentinelforge-api.kuinkel.com/api/v1/ingest/microsoft/windows-event` (or the endpoint matching the query)
Headers	`Authorization` : `Bearer @{body('Get_secret')?['value']}` `Content-Type` : `application/json`
Body	`@{body('Run_query_and_list_results')?['value']}`

Turn on **Secure inputs** for this action as well. If more than 500 rows can arrive in 5 minutes, run the trigger more often or add a *Chunk* step (batches of 500).

6 Save, click **Run**, and check the HTTP action's output: `{"received": …, "inserted": …, "skipped": …}`.

8. Any other source: the JSON ingestion API

Anything that can make an HTTPS request — a Linux server script, a SaaS webhook, a SIEM export, an EDR — can send events directly in SentinelForge's normalised format.

Request

```
POST https://sentinelforge-api.kuinkel.com/api/v1/ingest/events
Authorization: Bearer sfk_...
Content-Type: application/json

{
  "events": [
    {
      "event_time": "2026-10-27T09:15:00Z",
      "event_type": "auth.failure",
      "host": "web-01",
      "user_name": "j.doe",
      "src_ip": "203.0.113.45",
      "outcome": "failure",
      "message": "sshd: Failed password for j.doe",
      "raw": {"program": "sshd", "pid": 4123}
    }
  ],
  "run_detection": true
}
```

Response: `{"received": 1, "inserted": 1, "duplicates": 0, "alerts_created": 0, "alerts_updated": 0, "skipped": 0}`

Event types

event_type	Use for	Important fields
`auth.success` / `auth.failure`	Logins (VPN, SSH, web apps, AD)	user_name, src_ip, host
`process.start`	Process creation	host, process_name, command_line, parent_process, file_hash
`iam.user_created`	Account creation	target_user, user_name (who did it)
`iam.group_member_added`	Group / role membership added	target_user, group_name
`log.cleared`	Audit log cleared	host, user_name
`service.stopped`	Service stopped	host, process_name (service name)
`network.connection`	Firewall / flow records	src_ip, dst_ip, dst_port, bytes_out, host
`dns.query`	DNS lookups	domain, host or src_ip
`proxy.request`	Web / proxy requests	url, domain, user_name, host

`email.delivered`	Mail gateway deliveries	email_sender, email_subject, user_name (recipient), url, domain
`file.created`	File creation	host, file_hash, process_name
`security.alert`	Alerts from other security tools	outcome = severity (low/medium/high), message = title, host, user_name

Examples

curl (Linux / macOS)

```
curl -sS -X POST https://sentinelforge-api.kuinkel.com/api/v1/ingest/events \
  -H "Authorization: Bearer $SENTINELFORGE_INGEST_KEY" -H "Content-Type: application/json" \
  -d '{"events":[{"event_time":"2026-10-27T09:15:00Z","event_type":"auth.failure","host":"web-
01","user_name":"j.doe","src_ip":"203.0.113.45"}]}'
```

PowerShell

```
$body = @{ events = @(@{ event_time = (Get-Date).ToUniversalTime().ToString("o"); event_type =
"auth.failure"
                         host = "web-01"; user_name = "j.doe"; src_ip = "203.0.113.45" }) } |
ConvertTo-Json -Depth 5
Invoke-RestMethod -Method Post -Uri https://sentinelforge-api.kuinkel.com/api/v1/ingest/events `
  -Headers @{ Authorization = "Bearer $env:SENTINELFORGE_INGEST_KEY" } -ContentType
"application/json" -Body $body
```

Python

```
import os, json, urllib.request, datetime
event = {"event_time": datetime.datetime.now(datetime.UTC).isoformat(), "event_type":
"auth.failure",
         "host": "web-01", "user_name": "j.doe", "src_ip": "203.0.113.45"}
req = urllib.request.Request("https://sentinelforge-api.kuinkel.com/api/v1/ingest/events",
    data=json.dumps({"events": [event]}).encode(), method="POST",
    headers={"Authorization": f"Bearer {os.environ['SENTINELFORGE_INGEST_KEY']}", "Content-
Type": "application/json"})
print(urllib.request.urlopen(req, timeout=30).read().decode())
```

Rules

- 1–500 events per request, body up to 1 MiB, at most 60 requests per minute per data source.
- `event_time` must include a time zone (e.g. `Z`) and may not be more than 5 minutes in the future.
- Unknown fields are rejected; put extra vendor data in `raw` (any JSON object).
- Sending the same event twice is harmless: duplicates are detected and ignored.

9. Threat-intelligence providers

When an analyst investigates an IP, domain, URL or file hash, SentinelForge asks each configured provider. Provider API keys are set on the server (Vercel), never in the web interface.

Provider	Environment variable	Where to get the key
NepSec Intel	`NEPSECINTEL_API_KEY`	nepsecintel.kuinkel.com → **Workspace** → **API keys** (admin or Pro account). Key starts with `tl_`.
VirusTotal	`VIRUSTOTAL_API_KEY`	virustotal.com → your profile → **API key**
AbuseIPDB	`ABUSEIPDB_API_KEY`	abuseipdb.com → **Account** → **API**
AlienVault OTX	`OTX_API_KEY`	otx.alienvault.com → **Settings** → **OTX Key**

Connect NepSec Intel (recommended)

1. Sign in at **https://nepsecintel.kuinkel.com** with an administrator or Pro account.
2. Open **Workspace** → **API keys**, create a key named *SentinelForge*, and copy it (shown once).
3. In **vercel.com** open project **sentinelforge-api** → **Settings** → **Environment Variables** → **Add New**: Key `NEPSECINTEL_API_KEY`, Value = the key, environments *Production* and *Preview*, tick **Sensitive**, **Save**.
4. **Deployments** → the latest production deployment → ⋯ → **Redeploy**. New variables apply only after a redeploy.
5. In SentinelForge, **Settings** → **Threat intelligence integrations** now shows *NepSec Intel: Configured*. On **Threat intel**, investigate an indicator: the NepSec Intel card shows its risk score, top findings and an **Open full report** link.

How verdicts are shown.
A provider reporting nothing is shown as *unknown*, never as *safe*. Results are cached for 24 hours per provider; use **Refresh** to bypass the cache. The analyst's own verdict is kept separately and, when set to *malicious*, adds the indicator to the network detection watchlist.

10. Verify the connection

10.1 Health and events

1. **Settings** → **Data sources**: the source shows *Reporting* (an event in the last 24 hours) and a growing event count.
2. **Events**: filter by the data source; open an event to see the normalised fields and the original record under *raw*.
3. **Overview** → **Telemetry health** lists every source with its last event time.

10.2 Safe test alerts

These create harmless, clearly identifiable activity. Tell your team before running them, then close the resulting alerts as *Benign positive* with a note.

Source	Test	Expected alert
FortiGate SSL-VPN	From one outside IP, sign in to the SSL-VPN portal with a test account and a wrong password **five times**, then the correct password, within 10 minutes.	SF-AUTH-001 "Sign-in succeeded for <user> after 5 failures"
FortiGate IPS	From a lab machine, browse to the EICAR test file (eicar.org) through a policy with antivirus enabled.	SF-EXT-001 "FortiGate antivirus: malware EICAR…"
Windows	On a test server, five wrong passwords then a correct one for a test account (RDP or `runas`).	SF-AUTH-001

Windows	Create a test user and add it to a privileged test group within 30 minutes, then remove it.
	`net user sf-test P@ss-Test-2026! /add`
	`net localgroup Administrators sf-test /add`
		SF-IAM-001

Entra ID	Create a test user in Entra ID and assign it a privileged role (e.g. User Administrator), then remove it.	SF-IAM-001
Defender	Run Microsoft's documented Defender for Endpoint detection test on an onboarded device.	SF-EXT-001

No real network or account needed?
Settings → *Synthetic scenarios* → **Inject** writes labelled synthetic events through the same pipeline, to see detections end to end.

11. Troubleshooting

HTTP responses from the ingestion API

Code	Meaning	Fix
200	Accepted	Check `inserted`, `duplicates` and `skipped`. Many `skipped` means the records were not of a supported kind (see the connector's table).
401	Missing, wrong or revoked key	Check the environment variable (no spaces or quotes). Rotate the key and update the shipper.
413	Too many records or body over 1 MiB	Reduce the batch size (max 500).
422	Invalid events (generic API)	The response names the field: timestamp without time zone, unknown field, bad IP, unsupported `event_type`.
429	Rate limit (60 requests/minute per source)	Send larger, less frequent batches. Shippers retry automatically.
5xx	Server or database temporarily unavailable	Shippers retry. Check https://sentinelforge-api.kuinkel.com/api/v1/health.

By connector

Symptom	Check
FortiGate relay: `received=0`	FortiGate cannot reach the relay: `diagnose sniffer packet any 'port 5514' 4`; Windows/ufw firewall on the relay; correct server IP in `log syslogd setting`.
FortiGate relay: `rejected_source` > 0	The firewall sends from another IP: add it to `--allow` or set `source-ip`.
FortiGate: events but no traffic	Policies do not log allowed traffic, or the default `--traffic interesting` filtered normal HTTPS (expected).
Windows shipper: “sent 0 events”	Not run as Administrator / SYSTEM; auditing not enabled (`auditpol /get /category:*`); no matching events since the last run.
Windows: process alerts never fire	Command-line logging for 4688 is off (chapter 5.1).
Graph shipper: `HTTP 403 … Authorization_RequestDenied`	Admin consent not granted, or a permission is missing (chapter 6.1 step 3).
Graph shipper: sign-ins fail with a licence error	The tenant has no Entra ID P1/P2: use `--streams entra-audit,defender-alert`.
Graph shipper: `invalid_client`	The client secret expired or was copied as the secret ID instead of the Value.

Source shows *Stale*	No event for 24 hours: the shipper stopped (check its scheduled task / service) or the source is quiet.

12. Security checklist and reference

Security checklist

- ☐ One data source and key per log sender.
- ☐ Keys stored only in environment variables, protected files (chmod 600) or Key Vault.
- ☐ FortiGate relay restricted with `--allow` and a host firewall rule.
- ☐ Shippers run as SYSTEM / a dedicated service account, not a personal account.
- ☐ Graph app registration has only the three read permissions; secret expiry tracked.
- ☐ Keys rotated at least yearly and whenever staff with access leave.
- ☐ Unused sources revoked; mistaken ones deleted.
- ☐ Former staff removed under Settings → Users and roles.
- ☐ Provider keys (NepSec Intel, VirusTotal…) only in Vercel as *Sensitive*.

Limits

Limit	Value
Events / records per request	500
Request body	1 MiB
Requests per source	60 per minute
Future timestamps	max 5 minutes
Health "Reporting"	event within 24 h
Threat-intel cache	24 h per provider
Database (free tier)	0.5 GB

Event field reference (generic API)

Field	Type	Notes
`event_time` required	ISO 8601 with zone	When it happened, e.g. `2026-10-27T09:15:00Z`
`event_type` required	string	One of the types in chapter 8
`outcome`	string ≤ 32	success / failure / allowed / blocked; severity for `security.alert`
`host`	string ≤ 255	Hostname of the machine involved (creates or updates an asset)
`user_name`	string ≤ 255	Account acting or signing in
`src_ip`, `dst_ip`	IPv4 / IPv6	Validated
`dst_port`	0–65535	
`domain`, `url`	string	URL up to 2048 characters
`process_name`, `parent_process`	string	Executable names, e.g. `powershell.exe`
`command_line`	string ≤ 8192	Stored and shown as inert text
`file_hash`	string ≤ 128	MD5, SHA-1 or SHA-256

`target_user`, `group_name`	string	For IAM events
`email_sender`, `email_subject`	string	For `email.delivered`
`bytes_out`	integer ≥ 0	Bytes sent, for transfers
`message`	string ≤ 4000	Human-readable summary
`raw`	JSON object	Original vendor fields, kept for investigation

More detail in the repository: docs/connectors-fortigate.md, docs/connectors-microsoft.md, docs/api.md and docs/detection-coverage.md.